

Rubber-Band Dynamics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Valence Gap and Re-Alignment Analysis

PENPAL Project

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1 Overview

This notebook compares **rubber-band dynamics** across conditions using the exchange-aligned data structure.

The main quantity is the **valence gap**:

- `valence_gap = slot_1_valence - slot_2_valence`
- `gap_change = current_gap - previous_gap`

A negative slope in `gap_change ~ previous_gap` indicates a rubber-band effect: when the slots diverge, the next aligned exchange tends to pull them back together.

As in the updated valence notebook:

- `author_1 / author_2` are treated as **slot 1 / slot 2**
- the analysis is restricted to **complete exchanges**

- the temporal window uses `analysis_turn / turn_index`, not the old raw-turn logic

Primary cross-condition object. For each story i , fit the OLS

$$\text{gap_change}_{i,t} = \alpha_i + \beta_i \cdot \text{valence_gap_lag1}_{i,t} + \varepsilon_{i,t}$$

and define the per-story reversion strength as $R_i = -\beta_i$, so that larger positive R_i means a stronger pull back together. R_i is already label-invariant (the gap enters squared/reversal-style in the relationship). We test $R_i \sim \text{condition}$ with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon. The pooled LMM slopes are kept as supplemental.

Article-use note. This notebook tests a dyad-level regulation mechanism, not an agent-level asymmetry. If R_i does not differ by condition, the conclusion is that emotional re-convergence is a general co-writing dynamic rather than a Human-AI-specific asymmetry.

2 Load Data

Table 1: Loaded valence data for rubber-band analysis

condition	n_stories	n_turns
Human-AI	100	692
Human-Human	36	288
AI-AI	80	640

3 Preprocessing

```
## Analysis rows: 1404
## Stories: 216
```

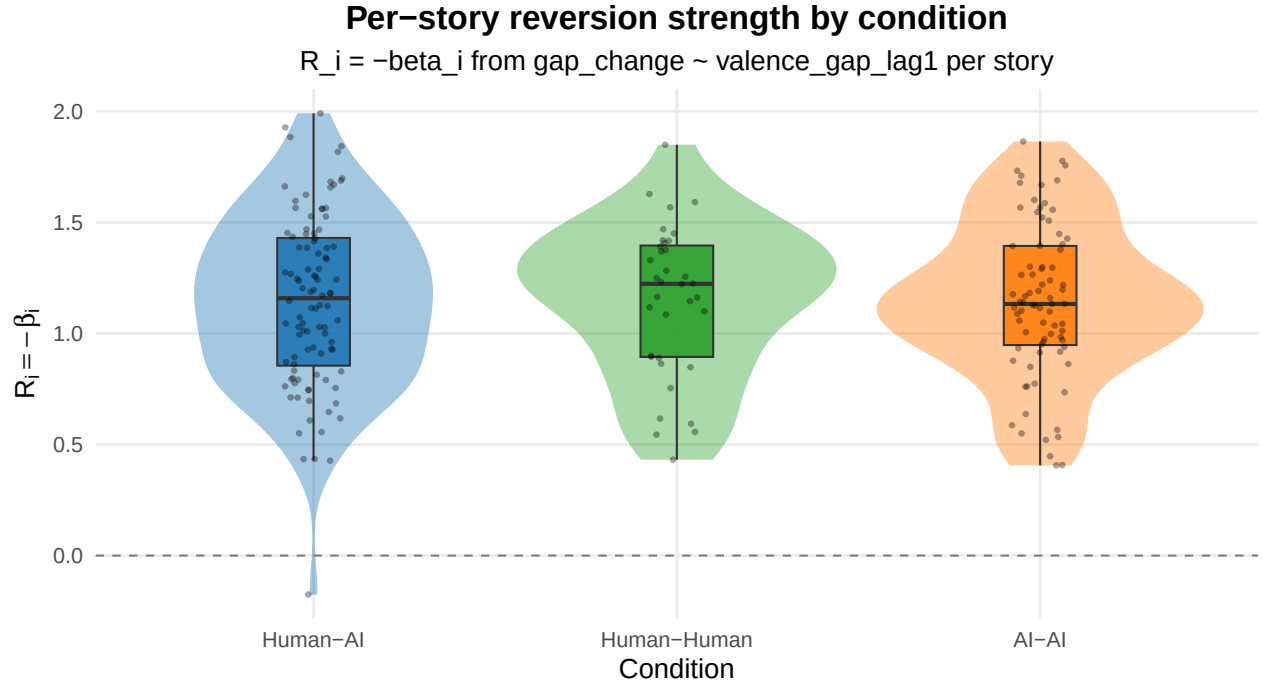
4 Primary: Per-Story Reversion Strength

Table 2: Per-story reversion strength $R_i = -\beta_i$ by condition

condition	n	mean_R	median_R	sd_R
Human-AI	100	1.144	1.159	0.388
Human-Human	36	1.150	1.224	0.342
AI-AI	80	1.145	1.133	0.349

```
## One-way ANOVA: R ~ condition
##           Df Sum Sq Mean Sq F value Pr(>F)
## condition    2  0.001  0.00045    0.003  0.997
## Residuals  213 28.671  0.13460
##
## Kruskal-Wallis (robust): R ~ condition
##
## Kruskal-Wallis rank sum test
##
## data:  R by condition
## Kruskal-Wallis chi-squared = 0.065776, df = 2, p-value = 0.9676
##
```

```
##
## Pairwise Wilcoxon (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: per_story_rb$R and per_story_rb$condition
##
##           Human-AI Human-Human
## Human-Human 0.99      -
## AI-AI       0.99      0.99
##
## P value adjustment method: BH
```



5 Bootstrap CIs and Effect Sizes on R_i

Table 3: Bootstrap 95% CI on per-condition mean of reversion strength R_i ($R = 10,000$)

condition	n	mean	ci_lower	ci_upper
Human-AI	100	1.144	1.068	1.218
Human-Human	36	1.150	1.038	1.256
AI-AI	80	1.145	1.068	1.220

Table 4: Pairwise Cliff's delta on R_i

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	100	36	-0.016	negligible
Human-AI vs AI-AI	100	80	-0.002	negligible

contrast	n_a	n_b	cliff_delta	magnitude
Human-Human vs AI-AI	36	80	0.041	negligible

6 Planned Contrast: HA vs (HH + AA)/2

Table 5: Planned contrast: R_i in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	-0.003	0.052	-0.06	167.092	0.953	-0.106	0.1

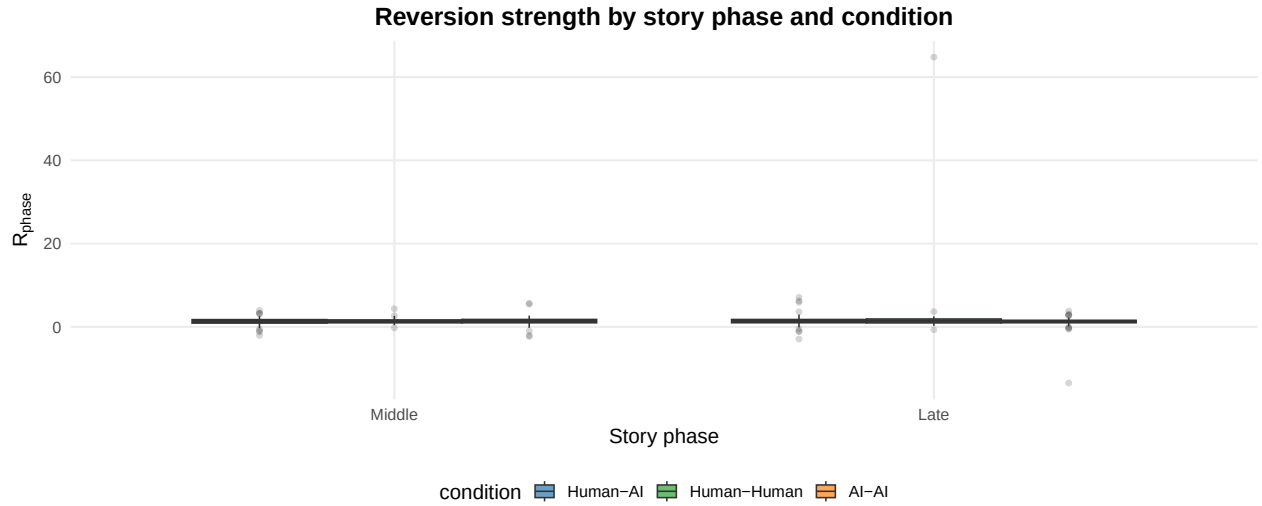
7 Time-Resolved Reversion Strength

Table 6: Per-story reversion strength by condition and story phase

condition	phase	n	mean_R	sd_R
Human-AI	Middle	94	1.281	0.927
Human-AI	Late	98	1.460	1.240
Human-Human	Middle	36	1.451	0.800
Human-Human	Late	36	3.135	10.598
AI-AI	Middle	80	1.363	1.108
AI-AI	Late	80	1.102	1.807

```
## LMM: R_phase ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: R_phase ~ condition * phase + (1 | conversation_id)
## Data: per_story_phase_rb
##
## REML criterion at convergence: 2208.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.4302 -0.1842  0.0044  0.1267 18.7107
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## conversation_id (Intercept)  0.00    0.000
## Residual                10.86    3.295
## Number of obs: 424, groups: conversation_id, 215
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    1.28074    0.33982 418.00000    3.769 0.000188
## conditionHuman-Human    0.17020    0.64577 418.00000    0.264 0.792246
## conditionAI-AI         0.08198    0.50117 418.00000    0.164 0.870148
## phaseLate         0.17902    0.47566 418.00000    0.376 0.706835
```

```
## conditionHuman-Human:phaseLate 1.50527 0.91067 418.00000 1.653 0.099096
## conditionAI-AI:phaseLate -0.43963 0.70543 418.00000 -0.623 0.533488
##
## (Intercept) ***
## conditionHuman-Human
## conditionAI-AI
## phaseLate
## conditionHuman-Human:phaseLate .
## conditionAI-AI:phaseLate
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cndH-H cndAI-AI phasLt cH-H:L
## cndtnHmn-Hm -0.526
## cndtnAI-AI -0.678 0.357
## phaseLate -0.714 0.376 0.484
## cndtnHm-H:L 0.373 -0.709 -0.253 -0.522
## cndtAI-AI:L 0.482 -0.254 -0.710 -0.674 0.352
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```



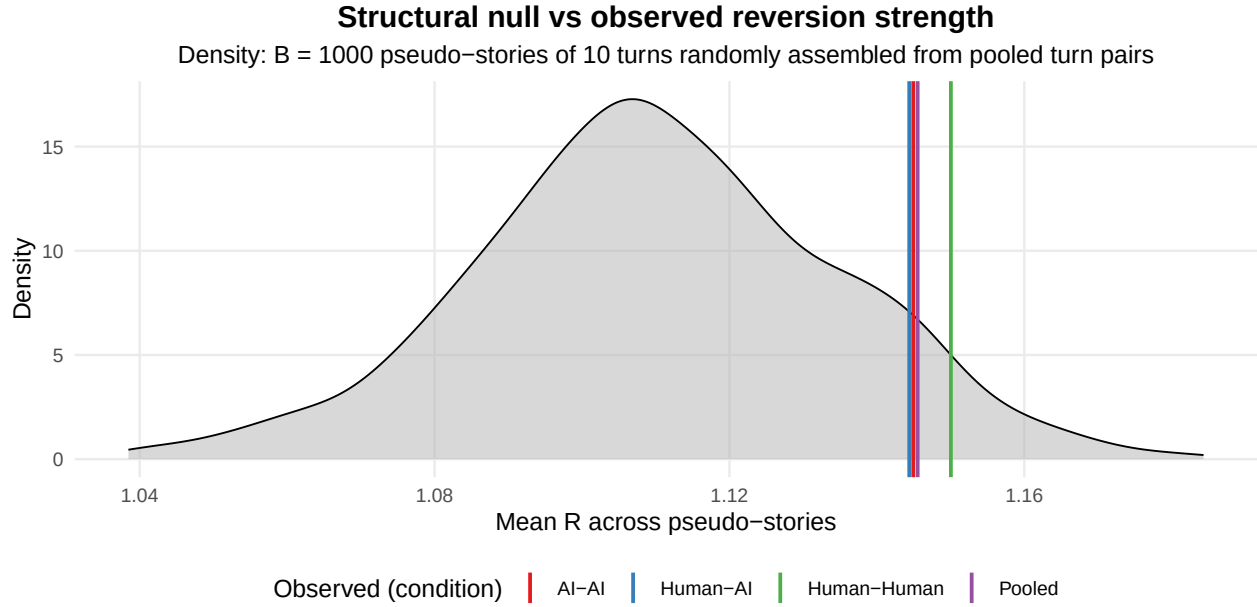
8 Structural-Null Permutation Test

Motivation: the observed rubber-band slope is near-identical across HH, HA, and AA, suggesting a structural / mean-reversion artifact of the `gap_change ~ gap_lag1` parameterization rather than genuine coordination. This test pools all turn-level (slot1, slot2) valence pairs across all stories and conditions, randomly repartitions them into pseudo-stories of 10 turns, and recomputes per-pseudo-story $R = -\beta$. If the observed mean R falls inside the resulting null distribution, the rubber-band effect is structural; if observed exceeds the null, it reflects genuine dyadic coordination.

Table 7: Structural-null test: observed mean R per condition vs 1000-iter null (pseudo-stories of 10 turns from 1620 pooled pairs)

condition	observed_mean_R	null_mean	null_sd	p_perm_two_sided
Human-AI	1.1444	1.11	0.0245	0.1668

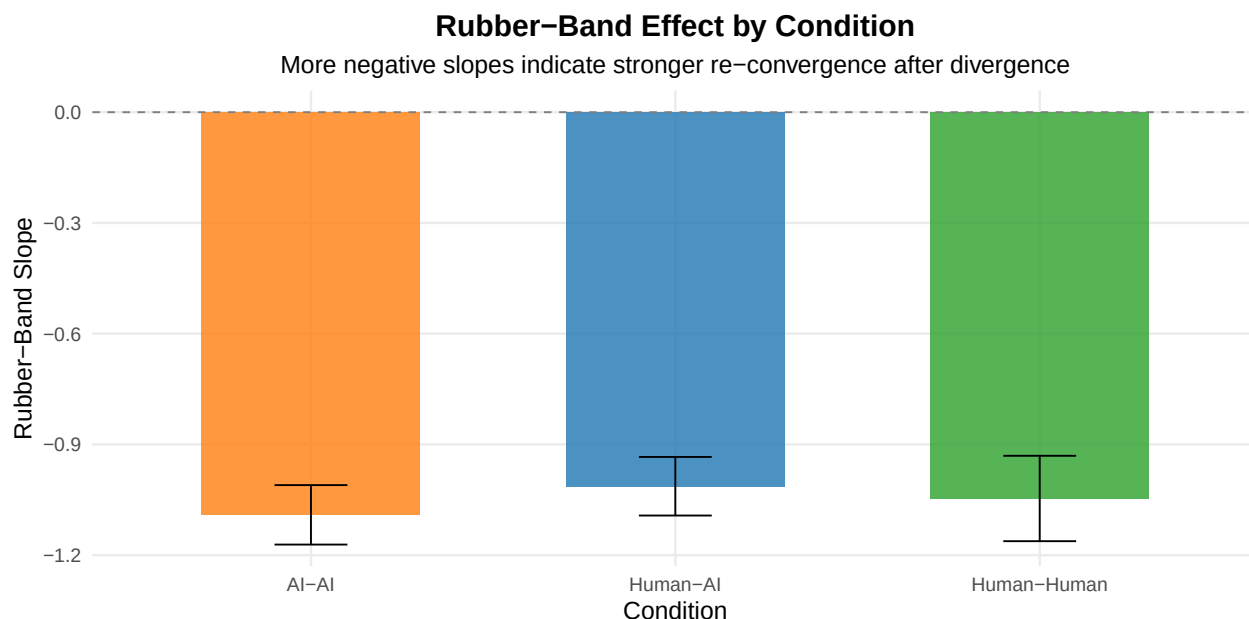
condition	observed_mean_R	null_mean	null_sd	p_perm_two_sided
Human-Human	1.1500	1.11	0.0245	0.0989
AI-AI	1.1449	1.11	0.0245	0.1588
Pooled	1.1455	1.11	0.0245	0.1518



9 Supplemental: Pooled Condition-Specific Slopes

Table 8: Rubber-band slopes by condition

condition	estimate	se	t	p	sig
Human-Human	-1.046	0.059	-17.733	0	***
AI-AI	-1.091	0.041	-26.545	0	***
Human-AI	-1.013	0.040	-25.043	0	***



9.1 Supplemental: Cross-Condition Interaction Model

```
## Cross-condition model summary:
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: gap_change ~ valence_gap_lag1 * condition + (1 | conversation_id)
## Data: df_analysis
##
## REML criterion at convergence: 1343.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.4228 -0.6365 -0.0155  0.6455  3.1225
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## conversation_id (Intercept) 0.009419 0.09705
## Residual                0.141790 0.37655
## Number of obs: 1404, groups: conversation_id, 216
##
## Fixed effects:
##              Estimate Std. Error    df t value
## (Intercept)   -0.10521   0.01870 186.11497  -5.627
## valence_gap_lag1 -0.96339   0.03731 1368.87498 -25.824
## conditionHuman-Human    0.10460   0.03426  147.67888   3.053
## conditionAI-AI         0.07960   0.02685  156.93525   2.964
## valence_gap_lag1:conditionHuman-Human -0.07174   0.06701 1394.60214  -1.071
## valence_gap_lag1:conditionAI-AI      -0.15859   0.06017 1388.90089  -2.636
##
##              Pr(>|t|)
## (Intercept)  6.64e-08 ***
## valence_gap_lag1 < 2e-16 ***
## conditionHuman-Human 0.00269 **
## conditionAI-AI      0.00351 **
```

```

## valence_gap_lag1:conditionHuman-Human  0.28457
## valence_gap_lag1:conditionAI-AI        0.00849 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) vln__1 cndH-H cAI-AI v__1:H
## vlnc_gp_lg1  0.212
## cndtnHmn-Hm -0.546 -0.116
## condtnAI-AI -0.696 -0.147  0.380
## vlnc__1:H-H -0.118 -0.557  0.053  0.082
## vl__1:AI-AI -0.131 -0.620  0.072  0.114  0.345
##
## Condition-specific slopes:
##   condition  valence_gap_lag1.trend      SE   df lower.CL upper.CL
## Human-AI      -0.963 0.0374 1378    -1.04   -0.890
## Human-Human    -1.035 0.0557 1381    -1.14   -0.926
## AI-AI          -1.122 0.0473 1353    -1.21   -1.029
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## Pairwise slope contrasts:
##   contrast                estimate      SE   df t.ratio p.value
## (Human-AI) - (Human-Human)  0.0717 0.0671 1396   1.069  0.8556
## (Human-AI) - (AI-AI)       0.1586 0.0602 1392   2.633  0.0257
## (Human-Human) - (AI-AI)     0.0869 0.0731 1371   1.189  0.7042
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests

```

10 Supplemental: General Gap Structure

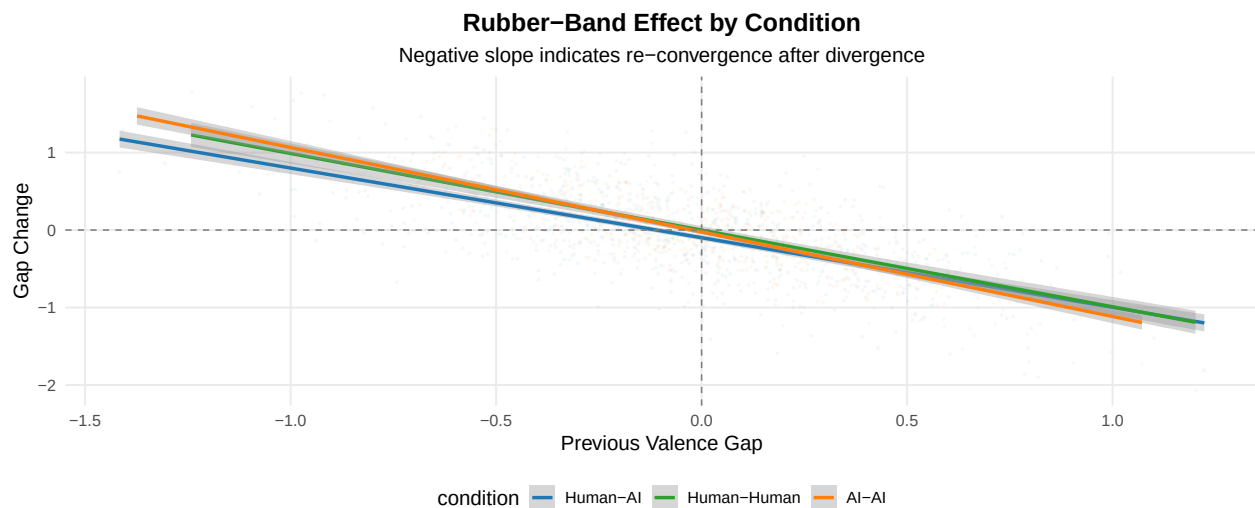
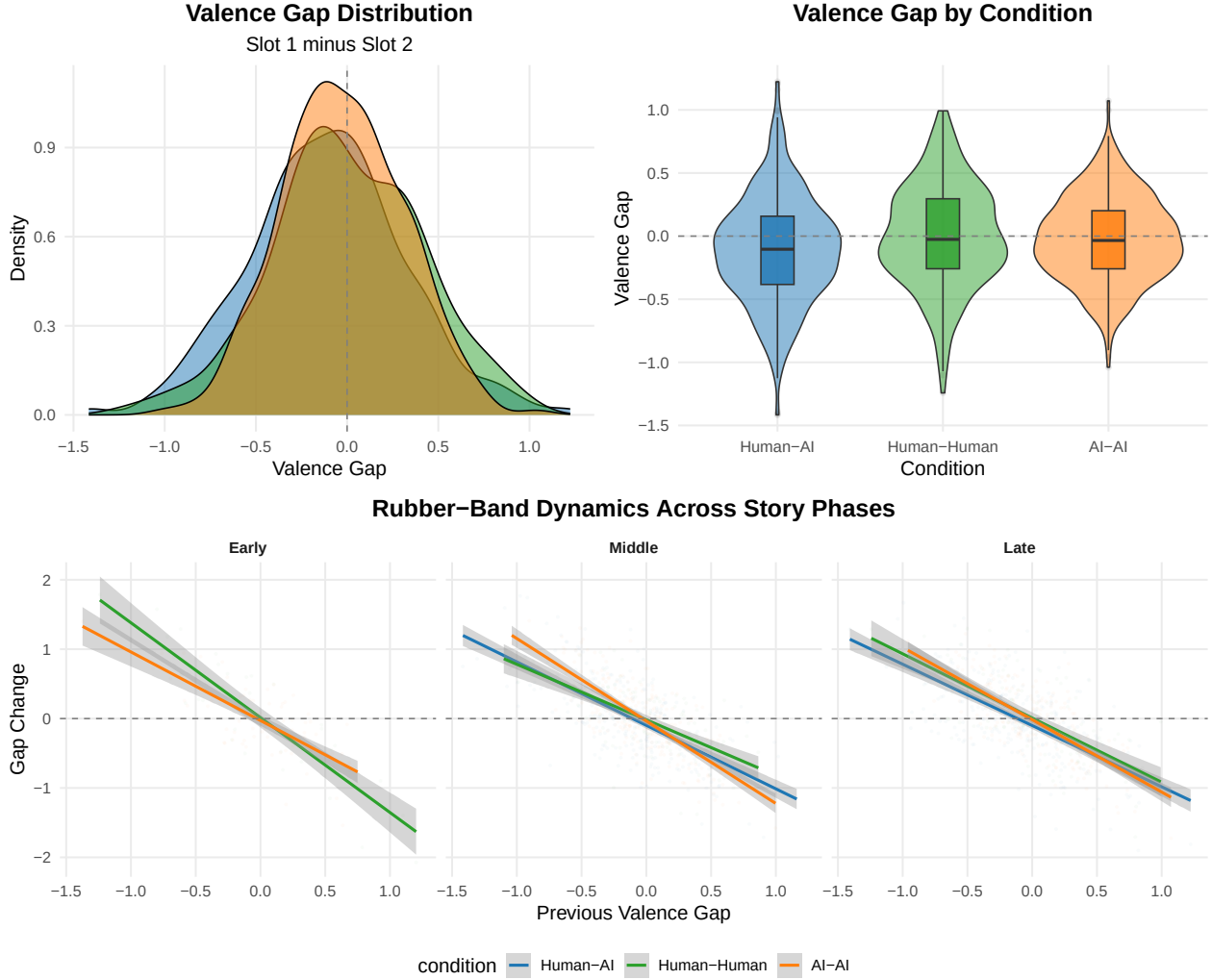


Table 9: Gap and gap-change descriptives by condition

condition	n_turns	mean_gap	sd_gap	mean_gap_change	sd_gap_change	rubber_band_r
Human-AI	592	-0.110	0.425	-0.003	0.574	-0.677
Human-Human	252	-0.001	0.413	-0.009	0.598	-0.724
AI-AI	560	-0.024	0.335	-0.007	0.502	-0.747



11 Summary

Primary cross-condition result: per-story reversion strength $R_i = -\beta_i$ from `gap_change ~ valence_gap_lag1`, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of R_i
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2
- time-resolved view of R across early / middle / late story phases

- **structural-null permutation test:** shuffles all turn-level valence pairs across stories and conditions, repartitions into pseudo-stories, and compares observed mean R to the resulting null. Tests whether the rubber-band effect is a coordination dynamic or a mathematical / mean-reversion artifact

Supplemental views retained:

- pooled within-condition LMM slopes (population-level rubber-band estimate per condition)
- cross-condition interaction LMM (`gap_change ~ valence_gap_lag1 * condition + (1|story)`)
- gap distributions and story-phase splits

The Human-AI rubber-band decomposition lives in `analysis/human-ai/rubber_band_analysis.Rmd` and is not duplicated here.

Deferred to a follow-up pass:

- starter-mechanism analysis of the reversion object

Analysis completed: 2026-05-05 12:49:11.307477